

More money, more effect? Employment effects of job search services of different intensity

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Abstract

Job search services modestly raise employment, but it is far from clear whether spending more on each participant improves outcomes further. We address this question by studying *Assegno per il Lavoro*, an Italian active labour market programme that assigns jobseekers to treatment groups with increasing endowments of job search services. Each intervention is made up of a voucher to fund job search assistance and a performance-based payment related to job search intermediation. Participants are assigned to groups with treatment endowments that increase according to a scoring variable. Leveraging this design, we apply a regression discontinuity analysis to estimate effects on both employment duration and employment quality. We find that providing a larger voucher for job search assistance does not increase take-up of the service, limiting its potential impact on employment. On employment duration, effects of higher endowments after 18 months are essentially null for both the average and the quartiles of days employed. On employment quality, we find that the intermediate endowment slightly increases the length of the contracts. Moreover, a design with different performance payments for job search intermediation seems to deter cream skimming, since workers are not discriminated in their access to the services. These results highlight both the risks of vouchers to implement public interventions and the potential of differentiated performance payments to deter cream skimming.

Keywords: active labour market policies, job search assistance, job search intermediation, public employment services, voucher

JEL codes: J64, J68, C21

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1 Introduction

Job search services are a type of active labour market policy (ALMP) usually qualified as effective to slightly increase the probability of employment (Card et al. 2018; Vooren et al. 2019). However, designing a programme like this requires knowledge of the specific tools and design features that optimize the average treatment effect. On the tools, recent research proposes to separate job search assistance from intermediation, suggesting that they influence the outcome through different mechanisms (Cheung et al. 2025). This distinction has not been addressed in previous meta-analyses, which have overlooked the differences between the assistance and intermediation mechanisms. In contrast, we analyze a programme that allows a particular focus on job search intermediation, a mechanism underexplored in the prior literature. On the design features, aspects like the intensity of the actions have been discussed when comparing different programmes. Literature reviews in economics (Vooren et al. 2019) and psychology (Liu et al. 2014) found that, in highly developed countries, the mean duration of an ALMP was not related to its employment effect. However, Escudero et al. (2019) detected the opposite for low and middle-income countries. This finding might suggest an interaction between the intensity of the intervention and other characteristics of the labour markets. If intensity effects are indeed null in developed countries, this would raise the question on the minimum treatment dose needed to generate a certain effect. Evidence of null intensity effects would justify reallocating resources toward covering more jobseekers rather than intensifying treatment for each participant.

This article studies the employment effects of different treatments with diverse doses of job search services. We evaluate an active labour market programme called *Assegno per il Lavoro* and implemented in Veneto (Italy) that offers different treatment endowments. Each participant of the treatment group is linked to a voucher of job search assistance and to services of job search intermediation. Importantly, both the voucher value and the intermediation payment increase across groups. Concerning the duration of employment, we study the effects of different treatment endowments on the mean and different quantiles of the distribution of days worked. Regarding employment quality, we study the effects on the probability of signing different types of employment contracts. Thanks to the assignment mechanism of the programme, we carry out a sharp regression discontinuity analysis that allows us to estimate local effects at the threshold. Identification was possible for two treatment comparisons: intermediate (B) vs. low endowment (A) and high (C) vs. intermediate endowment (B).

The results on the employment effects show that increasing the endowment is in general ineffective to increase the number of days worked, but it avoids the discrimination of hard-to-place jobseekers. In terms of employment duration, the impact on the average of days worked is eventually null at all semesters for the first treatment comparison (B vs. A) and of 10 extra days during the first year for the second treatment comparison (C vs. B). Looking at the median of the distributions, we estimate that there is a gain of employment in the intermediate-low contrast during the third semester and in the high-intermediate comparison during the second semester. The drawback is that all these effects vanish subsequently. In terms of employment quality, the

intermediate-low comparison triggers a higher probability of getting an open-ended contract during the implementation period of the programme. In contrast, this is not detected for the high-intermediate comparison, which generates an increase in the probability of signing a short temporary contract as the longest employment episode. Concerning programme implementation, on the one hand, we detect a very small change in the actual duration of job search assistance at the threshold, consistent with prior evidence on implementation failures in voucher-based public services. Consequently, job search intermediation appears to be responsible for the few impacts observed. On the other hand, under the differentiated payment structure, we find no evidence of discrimination in access to job search services for the less-employable jobseekers.

Related literature. There is a broad literature estimating the effect of participating in job search services on employment attainment both in economics (Card et al. 2018; Vooren et al. 2019) and psychology (Liu et al. 2014). Our work contributes to three research lines that deal with the employment effects of the following design features: the intensity of assistance, the intensity of intermediation, and the private provision of public employment services.

Firstly, job search assistance (JSA) is aimed at improving the job search of the unemployed individual by intervening on her job search skills, motivation, or targets. The heterogeneous effects of JSA in function of its intensity have been rarely studied in an European context. Liu et al. (2014) performed a meta-analysis on the employment effects of JSA including 32 randomized experiments and 15 observational studies, most of them based on the USA. They concluded that there was not an association between the intensity (duration) of the intervention and the effect magnitude. Martinson et al. (2019) and Dyke et al. (2006) also found a null effect of JSA intensity on employment with studies almost exclusively from the USA. On the contrary, Meyer (1995) detected a positive employment effect also in the USA. More research is needed for the European countries, as there are substantial differences in the organization of public administrations and the markets for ALMPs providers between the USA and Europe.

Secondly, job search intermediation (JSI) involves a delegation of the job search from the worker to a third party, whether as a full delegation or as a complementary service. There are only a few evaluations of these interventions, but they usually conclude positive effects on employment (Fougère et al. 2009; Engström et al. 2012; Van Den Berg et al. 2019; Cheung et al. 2025). Three other articles have investigated the effect of JSI under performance-based payments. Zanella and Salomone (2025) found a positive impact in private providers, whereas Koning and Heinrich (2013) concluded the same under a mix of ALMPs but only for non-disable jobseekers. Burgess et al. (2017) studied public providers and only detected positive effects in large establishments.

Thirdly, researchers have also studied the potential benefits and risks of a private provision of public employment services via vouchers and performance pays (Hipp and Warner 2008). Regarding a provision with vouchers, in the American context, Barnow (2009) reviewed evaluations of voucher implementations in adult training, noting mixed estimates on the employment effects. Doerr et al. (2017) found that, on average, receiving a voucher generated large and lasting negative effects on employment. Like Huber et al. (2018), they found small positive effects emerging in the very long run. In the European context, Brenninkmeijer and Blonk (2012) conducted an experiment

on JSA in the Netherlands, evaluating an open-dose (voucher) against a closed-dose treatment and a no-treatment group. In the short term, the voucher condition yielded results similar to the control condition. An Italian programme evaluated by Linfante et al. (2019) found null effects on employment one year after treatment for compliers who registered after receiving an invitation.

Concerning a provision with performance pays, one of the potential disadvantages of a quasi-marketization of public services is the cream-skimming problem. Cream-skimming refers to the rejection of hard-to-place jobseekers by service providers. The evidence on the existence of this problem is mixed, with some studies finding it for certain groups (Koning and Heinrich 2013), other studies rejecting its existence (Aakvik et al. 2005) and other articles highlighting the measurement problems of this concept (Heckman and Smith 2004). Establishing different performance pays in function of the jobseeker’s employability has been sometimes presented as a shelter against this problem.

This paper makes three contributions to the literature. First, we provide new evidence on the dose-response function in job search assistance within a European context, where programme designs and labour market institutions differ substantially from the US-based prior literature. Second, we study the heterogeneous effects of job search intermediation under performance-based payments, focusing for the first time on how the intensity of such payments shapes employment outcomes. Third, we assess whether differentiated performance payments succeed in discouraging cream-skimming, contributing to the literature on the design of quasi-markets for public employment services. On the methodological side, our analysis uses a credible identification that combines regression discontinuity with local quantile treatment effects, one of the first applications of these recently proposed techniques.

We continue in Section 2 describing the programme design evaluated. In Section 3, we present the data and the methods employed to answer our research questions. Section 4 shows the results of our work and assesses the plausibility of our identification assumptions. Lastly, we draw some conclusions in Section 5.

2 Policy design

Assegno per il Lavoro (hereinafter, AXL) was a programme developed in Veneto (Italy) from July 2017 to May 2022 and funded by the public employment office of the region. The general objective of AXL was to reduce unemployment. Initially, the target group were those over 35 years old that met at least one of the conditions of social vulnerability included in Government of Veneto (2017a). From April 2018, the programme extended to all unemployed individuals over 30 years old.

To achieve this objective, policymakers designed an intervention with diverse tools or components. The individual could receive job search assistance, adult training, job search intermediation, and internships. The intensity of such intervention could vary according to the treatment group. Table 1 shows the amount of money available for each participant related to each component by treatment group. There are three main sources of variation between groups. Regarding job search assistance, group *B* almost doubles the number of maximum hours available compared to

group *A*. We can see a remarkable difference on this matter between group *C* and group *B* as well. Concerning job search intermediation and internships, the money devoted to such components also changes by treatment group. Lastly, the maximum duration of the intervention also increases from *A* to *C*, although the differences are not substantial.

Table 1: Endowments and duration of AXL by treatment group ([Government of Veneto 2017a](#); [Government of Veneto 2018](#))

	Group A	Group B	Group C
Total endowment	3,536 €	4,264 €	5,796 €
- Job search assistance	266 € (7 hours)	494 € (13 hours)	1,026 € (27 hours)
- Adult training	1,770 €	1,770 €	1,770 €
- Job search intermediation*	1,500 €	2,000 €	3,000 € (for JSI and internships)
- Internships*	0 €	0 €	
Maximum duration	3 months (6 with interruption)	5 months (8 with interruption)	6 months (9 with interruption)

The implementation process is carried out through vouchers and performance-based pays. The individual decides first where to spend it. There is a quasi-market of private providers that offer all these services. The public employment office lists all of them and the individual chooses where to go. On the voucher side, participant and provider decide then how much to spend on job search assistance and on adult training. The hourly price of the first component is fixed, but it is not for the second one. On the prize side, job search intermediation and internships are not paid depending on the service provision. It is strictly a performance-based payment: the provider only gets the money if the participant signs an employment contract. Moreover, the performance-based pay increases in function of the duration of the contract¹.

Table 1 shows that the endowment of the programme depends on the treatment group in which the participant is allocated. In AXL, the individual is assigned to a certain treatment group according to a scoring variable that estimates the probability of being employed in 24 months. The score has the following functional form ([Government of Veneto 2017b](#)):

$$\hat{S}_i = 1/[1 + \exp(-(\hat{\beta}_0 + \hat{\beta}_1 sex_i + \hat{\beta}_2 age_i + \hat{\beta}_3 foreign_i + \hat{\beta}_4 studies_i + \hat{\beta}_5 disab_i + \hat{\beta}_6 termination_i + \hat{\beta}_7 occup_i + \hat{\beta}_8 sector_i))]$$

$\hat{S}_i$ incorporates both sociodemographic covariates (sex, age, being foreign, level of studies, and having a disability) and covariates on the last employment (reason of termination, major group of occupation, and sector). To make the scoring variable following the same direction of treatment

¹Table 1 shows the performance-based pay for JSI in case of an open-ended contract. If a fixed-term contract of more than 12 months is signed, this quantity is multiplied by 0.8. If the duration is between 6 and 12 months, the factor is 0.6.

endowment, we reversed the original scoring variable generating S_i . Once the participant receives a score, it is allocated to a certain treatment group according to the next function:

$$group_i = \begin{cases} A & 0 \leq s \leq 0.4 \\ B & 0.41 \leq s \leq 0.6 \\ C & 0.61 \leq s \leq 1 \end{cases}$$

3 Methods

3.1 Data

The data comes from administrative records of Veneto Lavoro, the public employment service of the region of Veneto (Italy). We use microdata on all individuals that applied to AXL during the whole implementation period of the programme.

We have information on three outcome variables measured at different periods. Firstly, related to the employment duration, we know the number of days worked during the first, the second, the third, and the fourth semester since the treatment started. Secondly, regarding employment quality, we have a polytomous outcome variable that registers what was the longest employment contract signed since the treatment began. Notice that the type of contract is considered one of the two dimensions of employment quality by Muñoz De Bustillo et al. (2011). We distinguished five categories: open-ended contract (OEC), temporary contract of long duration (i.e. more than 12 months), temporary contract of intermediate duration (i.e. from 6 to 12 months), temporary contract of short duration (5 or less months), and no employment. We also have information about the type of contract linked to the performance-base pay of intermediation. In some cases, such contract may be the same as the longest employment contract, but not necessarily. Thirdly, related to treatment uptake, we know the doses of treatment actually received by participants. We have the number of hours of job search assistance and the number of hours of adult training. This last set of outcomes will help us disentangle some possible mechanisms to explain the results.

The remaining covariates in our dataset are the scoring variable and other sociodemographic variables. The scoring or running variable is the unique confounder according to the programme design, so we do not need additional control variables for identification. The remaining sociodemographic covariates will be employed for the plausibility and heterogeneity analyses. In the supplementary document, we include the full list of variables with their definitions.

3.2 Analysis

3.2.1 Target parameters

We consider different outcomes, so we denote as Y_{oi} the response variable for outcome o and individual i . For a simpler exposition, we remove the subindex o without loss of generality. During the presentation of results, we also eliminate subindex i to facilitate the reading. We denote the

potential outcome for the state of the world with the active treatment as $Y_i(1)$ and the potential outcome for the state of the world without that active treatment as $Y_i(0)$.

We are interested in two treatment comparisons. D_{1i} is a binary variable that collects if individual was treated under group B (with value 1) or under group A (with value 0). D_{2i} is another binary variable that informs if individual was treated under group C (with value 1) or under group B (with value 0). For each case, treatment with value 1 is referred as the active treatment, whereas treatment with value 0 is called the control treatment. As we mentioned before, treatment group is determined by the scoring variable S_i . For each treatment comparison, we partition the sample by delimiting the range of the score and centring it at the cutoff, so both S_{1i} (affecting D_{1i}) and S_{2i} (affecting D_{2i}) present the cutoff at $c = 0$. Therefore, we may denote more generally D_{zi} and S_{zi} for $z \in \{1, 2\}$, but in the following we omit subindex z to reduce the notational burden.

Our target parameters are the local average treatment effect of D_i on Y_i ($LATE$), the local average treatment effect of D_i on the probability of $Y_i = j$ ($LATE_j$), and the local quantile treatment effect for a certain quantile Q_τ at percentile τ of D_i on Y_i ($LQTE(\tau)$). In all cases, by local we mean the subpopulation with a score just at the threshold, which determines receiving the higher endowment in each comparison. Equivalently, these target parameters can be understood as local effects of a certain endowment or as intention-to-treat effects of a certain treatment. Formally, we can write the estimands as

$$LATE = \mathbb{E}[Y_i(1)|S_{zi} = c] - \mathbb{E}[Y_i(0)|S_{zi} = c] \text{ for } z \in \{1, 2\}$$

$$LATE_j = \mathbb{P}[Y_i(1) = j|S_{zi} = c] - \mathbb{P}[Y_i(0) = j|S_{zi} = c] \text{ for } z \in \{1, 2\}$$

$$LQTE(\tau) = Q_\tau[Y_i(1)|S_{zi} = c] - Q_\tau[Y_i(0)|S_{zi} = c] \text{ for } z \in \{1, 2\}$$

3.2.2 Identification

Since the target parameters contain unobservable potential outcomes, we require identification assumptions to estimate them using observable potential outcomes. Following Rubin (2007), an identification strategy is more credible when it is closer to the properties of a randomized experiment. Therefore, we propose to leverage the design of AXL to use a sharp regression discontinuity strategy. This method is based on two identification assumptions related to two properties of randomized experiments.

Assumption 1

The assignment mechanism is $D_i = \mathbf{1}(S_i \geq c)$, so $\lim_{s \downarrow c} \mathbb{E}[D_i|S_i = s] - \lim_{s \uparrow c} \mathbb{E}[D_i|S_i = s] \neq 0$.

The first identification assumption is the unconfoundedness of the assignment mechanism, which departs from the knowledge on how the treatments were assigned (Abadie and Cattaneo 2018; Frölich and Sperlich 2019). The main idea is that adjusting for the scoring variable, the assignment variable is not related to any potential outcome. In our case, it is a weak assumption thanks to the detailed knowledge on the treatment assignment function. Both the awareness of the

assignment function and the unconfoundedness of the assignment mechanism are properties of randomized experiments.

If treatment assignment was not only influenced by the score, but also by other unadjusted variables like ability, Assumption 1 would be false. This problem is known as score manipulation (Frölich and Sperlich 2019), so we will examine how feasible is to manipulate the specific variables that determine the score used in AXL.

Assumption 2

- For $LATE$: $\mathbb{E}[Y_i(1)|S_i]$ is continuous at $S_i = c$ and $\mathbb{E}[Y_i(0)|S_i]$ is continuous at $S_i = c$.
- For $LATE_j$: $\mathbb{P}[Y_i(1) = j|S_i]$ is continuous at $S_i = c$ and $\mathbb{P}[Y_i(0) = j|S_i]$ is continuous at $S_i = c$.
- For $LQTE(\tau)$: $F_{Y(1)|S}(y|c)$ is continuous and $F_{Y(0)|S}(y|c)$ is continuous.

The second identification assumption solves the absence of overlap through the continuity of potential outcome functions (Frölich and Sperlich 2019). While the designs of randomized experiments present probabilistic assignment mechanisms, the designs linked to the sharp regression discontinuity are based on a deterministic assignment mechanism. There are two approaches to bypass this problem in the regression discontinuity context (Cattaneo et al. 2019). The local randomization approach is suitable when we have a natural experiment as defined by Titiunik (2021). It is not our case, because our assignment mechanism does not incorporate any element of chance. The continuity approach to regression discontinuity solves the lack of overlap by performing a small extrapolation. The rationale is that extrapolations distant from the observed data are more model-dependent (King and Zeng 2006), so this strategy focuses on a more modest estimand around the cutoff.

There are two situations that might provoke violations of Assumption 2. On the one hand, if another programme employed the same scoring variable to allocate treatments and its effect was not null. Thereby, $\mathbb{E}[Y_i(0)|S_i]$ or its homologous would suffer a discontinuity at the cutoff, because this would be the point at which the alternative treatment is activated. Two checks will be performed to assess the plausibility of this situation. First, we will use pseudo-treatment or placebo outcome tests just at the cutoff (Frölich and Sperlich 2019; Eggers et al. 2024). Following Manski and Pepper (2018), a statistically significant difference between groups would diminish the credibility of the continuity assumption. Second, we will examine the documents of the AXL design to judge the possibility of other programmes employing the same running variable. On the other hand, if treatment groups in our sample had different conditional expected values in covariates included in the confounding variable, the credibility of the continuity assumption would also be lower. Therefore, we will study the balance of these covariates among the different treatment groups.

3.2.3 Estimation and inference

We are going to estimate the target parameters using graphical and quantitative methods. As a first step, we use regression discontinuity plots both with local mean/proportions and with local medians (Cattaneo et al. 2024). As a second step, we analyse the data with more fine-grained estimation and inference techniques.

Regarding point estimation, we have used three estimators, one for each estimand. For quantitative outcomes, we can credibly estimate both the $LATE$ and the $LQTE(\tau)$. In order to estimate the $LATE$, we have used the local polynomial estimator Cattaneo and Titiunik (2022). A common bandwidth for both sides has been chosen by minimizing the mean squared error following standard procedures (Calonico et al. 2017). For each $LATE$, two specifications have been estimated, one quadratic and other linear in the score, both with the triangular kernel². The definitive specification has been selected according to the asymptotic mean squared error as recommended by Pei et al. (2022).

To estimate the $LQTE(\tau)$, we have applied the local estimator proposed by Qu and Yoon (2019). The procedure to choose the bandwidth is adapted to the quantile setting but preserves the same intuition: the minimization of the mean squared error. We have estimated the optimal bandwidths according to three procedures proposed by Qu and Yoon (2015) and Qu and Yoon (2019): treating the cutoff as an interior point, as a boundary point, and with the quantile adaptation of the bandwidth selector by Imbens and Kalyanaraman (2012). Then, to avoid bias, we have selected the lowest bandwidth among the three alternatives. As recommended by the authors of the method, all specifications consider a linear specification of the regression coefficients of S_i and employ the Epanechnikov kernel. For quantile regressions, we do not maintain the rank invariance assumption, so interpretation of results must be made in accordance with that. Under rank dissimilarity, estimates of $LQTE(\tau)$ are not informative about on which individual the treatment is most effective (Dong and Shen 2018). Nonetheless, they inform about changes at the potential outcome distributions, which helps to assess the usefulness of the programme as a whole.

For qualitative outcomes, we can only estimate the $LATE_j$. We have employed the local logit model with maximum likelihood estimator developed by Xu (2017). The bandwidth has also been chosen by minimizing the mean squared error, with a procedure adapted to the categorical setting by Xu (2017). His method also departs from a linear specification of the coefficients of S_i and uses a uniform kernel.

Concerning inference, methodologists have considered that non-parametric estimators minimize bias *and* variance when optimizing the mean squared error, so some bias remains. Following Cattaneo et al. (2019), we use a bandwidth that is optimal for point estimation but removing the remaining bias for inference aims. For the mean estimand, we executed a Wald test of no effect employing the robust bias-corrected approach of Calonico et al. (2014). For the probability estimand, Xu (2017) adapts the approach of Calonico et al. (2014) to the categorical setting also with a Wald

²We chose the triangular kernel because of its point estimation optimality property, its consistency in spirit with the available kernels of Qu and Yoon (2019) and its compatibility with the procedure of Pei et al. (2022). The results are essentially the same with a uniform kernel for almost all semesters.

test. For the quantile estimand, the first inference procedure contrasts treatment non-significance and is a Wald test of $H_0 : LQTE(\tau) = 0$ for all $\tau \in \mathcal{T}$, with $\mathcal{T}$ denoting the quantile range, and $H_1 : LQTE(\tau) \neq 0$ for some $\tau \in \mathcal{T}$. The second inference procedure contrasts the homogeneity of treatment effects across quantiles and is a Wald test of $H_0 : LQTE(\tau)$ is constant over $\tau \in \mathcal{T}$, with $H_1 : LQTE(\tau) \neq LQTE(v)$ for some $\tau, v \in \mathcal{T}$. These tests developed by Qu and Yoon (2019) are also robust-bias-corrected in the sense of Calonico et al. (2014).

To assess the robustness of the results, we will repeat the analyses with variations in modelling options such as the polynomial degree, slight changes of the optimal common bandwidth and different optimal bandwidths at each side of the cutoff. Additionally, we will also run subgroup analyses to look for heterogeneous effects. This strategy repeats the analyses in strata of the population formed by covariates that might be an effect modifier of the treatment.

4 Results

4.1 Effects on assistance take-up

The design of AXL allows us to identify the effect of different treatment endowments on actual treatment uptake. It seems reasonable to assume that policymakers give more money to the less-employable people to encourage them to take a more intense treatment. We can measure the intensity of such treatment through the number of hours actually spent in job search assistance (Y_{HS}).

Figure 1 displays visual evidence on the local means of hours of treatment uptake for each endowment in function of the score of the participant. Concerning the first treatment (D_1), we appreciate a jump at the cutoff for the actual hours of job search assistance. Regarding the second treatment (D_2), the number of hours of job search assistance also presents a small change of level at the cutoff, although much smaller than the potential difference allowed by the endowment.

Figure 1: RD plots for the effect on assistance take-up. LSM: Local Sample Mean.

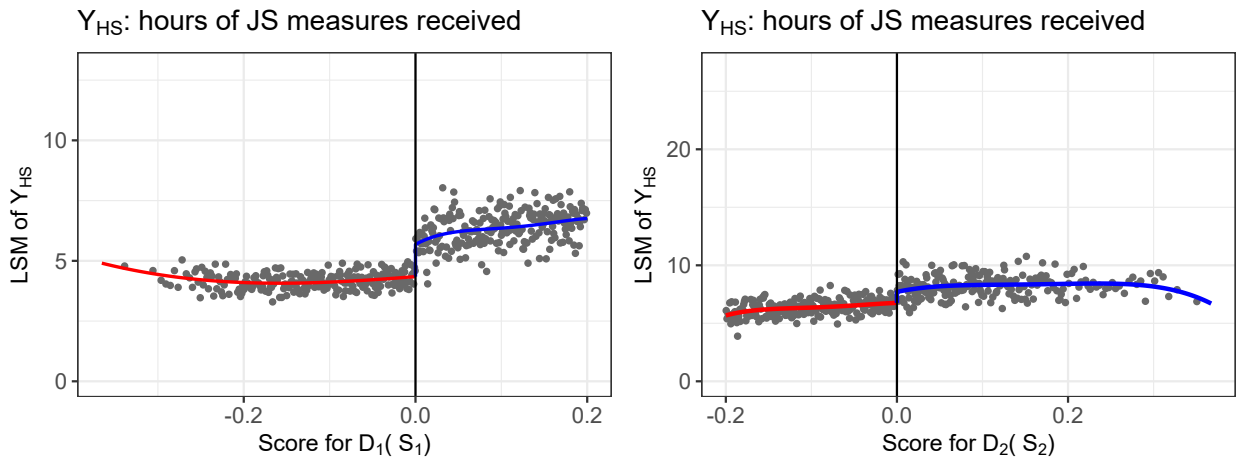


Table 2: Estimates of effects for each treatment assignment at the mean of hours of assistance take-up.

	Y_{HS}
D_1	1.255*** (0.210)
Bandwidth (h)	0.046
n_h	4774
D_2	1.098*** (0.255)
Bandwidth (h)	0.074
n_h	8013

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each cell shows the point estimate of the $LATE$ and its standard error in parenthesis. n_h is the effective sample size. The polynomial degree is always one.

To complement the plots with quantitative estimates of the effects, Table 2 offers formal estimates for each treatment. All results are consistent with the graphical evidence. Firstly, we find a mean difference of only an extra hour of job search assistance in the active treatment both for D_1 and D_2 . This difference is statistically significant at the 1 % level in both cases. The magnitude is remarkably low once we consider that the potential difference between both groups is 6 hours (in D_1) and 14 hours (in D_2). As we show in the supplementary document, the different amount of the vouchers for job search assistance does not have a statistically significant effect on the number of training hours either. This is less surprising due to the equal amount of money reserved for training in every group. Therefore, considering the small differences of participation in assistance between control and active treatments, the results below may be interpreted as the unique effect of different performance pays for job search intermediation.

4.2 Effects on employment

Figure 2 presents regression discontinuity plots with local means of the number of days worked at the first and second semester for the first (score S_1) and the second treatment (score S_2). Plots for the remaining semesters are available in the supplementary document. We see that there is no jump at the cutoff in any case, which suggests a null effect of both the first increase (D_1) and the second increase in the treatment dose (D_2). Although the visual evidence offers a negative evaluation of the interventions, more formal methods are required to quantify such differences.

Table 3 offers formal estimates of the $LATE$ for both treatments, with results that are consistent with those presented in previous plots. The first panel shows the mean effect of D_1 at the cutoff across the four semesters. We do not have evidence of any statistically significant average effect at

Figure 2: RD plots with local means for employment duration variables. LSM: local sample mean.

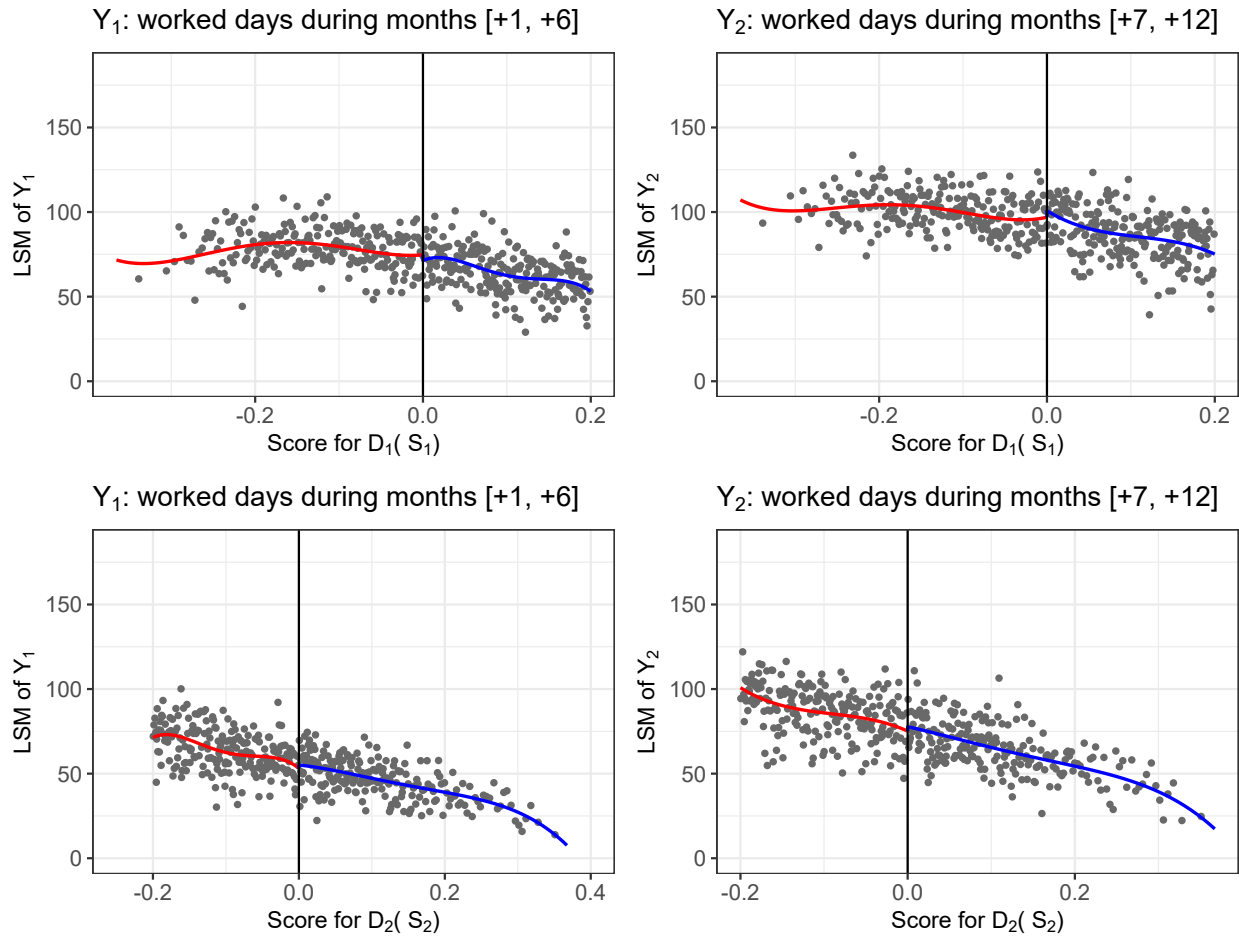


Table 3: Estimates of employment duration effects for each treatment at the mean, semesters 1 to 4.

	Y_1	Y_2	Y_3	Y_4
D_1	-1.890 (3.698)	4.739 (4.183)	7.961 (5.115)	4.428 (4.743)
Bandwidth (h)	0.064	0.061	0.045	0.053
n_h	6634	6375	4716	5495
D_2	6.980* (4.383)	10.141** (4.867)	4.229 (4.752)	7.178 (5.256)
Bandwidth (h)	0.037	0.044	0.049	0.044
n_h	4077	4885	5309	4823

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each cell shows the point estimate of the $LATE$ and its standard error in parenthesis. n_h is the effective sample size. The polynomial degree is always one.

any of the semesters analysed. Setting aside statistical inference, the point estimates are uniformly small, ranging from an average decrease of one to an average increase of eight days worked. The second panel presents the mean effect of D_2 at the cutoff for the same periods. Considering that the difference between active and control treatments is larger in this case, the impact would be larger under a linear dose–response function. Our findings indicate that the effect is only statistically significant at the second semester (for $\alpha = 0.05$), with a moderate effect magnitude of ten days worked. During the rest of the semesters, the effects are not statistically significant at the conventional significance level. Therefore, at least for those at the threshold, our analyses suggest that the effect of both increases in the treatment dose is virtually null for most semesters³.

Even if the treatments caused a null effect at the mean of our outcome variables, it is possible that they shifted the location of certain quantiles. Figure 3 explores the local medians of both treatments before and after the threshold for semesters 1 and 2. Figures for the rest of semesters and for the effects at other quantiles are available in the supplementary document. For the first active treatment (D_1), we barely appreciate a discontinuity at the cutoff for the number of days worked at semester one or two. For the second active treatment (D_2), the situation is quite similar if we focus on the first semester. The figure for the second semester presents a more pronounced jump at the median of days worked just at the threshold. This might be suggesting a positive effect of this treatment on the employment duration.

Table 4 shows estimates of the $LQTE(\tau)$ on the employment duration of each semester, that is, the effect on the different quartiles of the distribution of days worked. As expected, the results are consistent with the visual evidence presented just before for the median. Regarding the first treatment (D_1), the first main result is the point estimate of zero effect on the first quartile for

³The results are robust to using “being employed at quarter t ” as alternative outcome. The estimates on the increases in the employment probability are still tiny (overall lower than 2 p.p. in absolute terms) and not statistically significant in all quarters.

Figure 3: RD plots with local medians for employment duration variables. LSMed: local sample median.

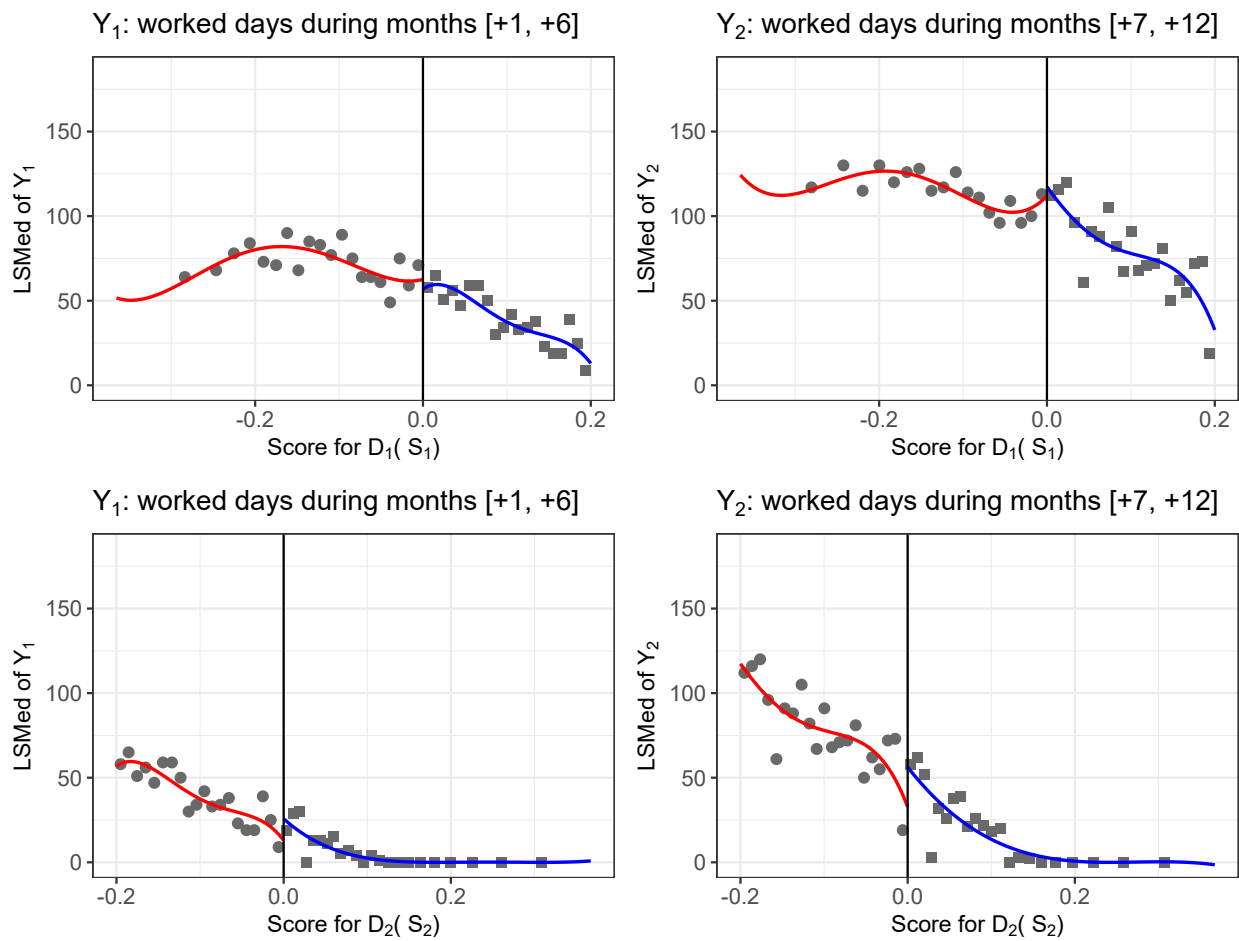


Table 4: Estimates of employment duration effects for each treatment at quantiles, semesters 1 to 4.

	Y_1	Y_2	Y_3	Y_4
D_1				
$\tau = 0.25$	0	0	0	0
$\tau = 0.5$	-5.072	10.555	21.91	12.322
$\tau = 0.75$	3.863	1.015	1.014	1
Treatment non-significance test			**	
Treatment homogeneity test				
D_2				
$\tau = 0.25$	0	0	0	0
$\tau = 0.5$	12.825	58.384	-7.183	-1.388
$\tau = 0.75$	-1.75	6.001	0	0
Treatment non-significance test		*		*
Treatment homogeneity test				

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. For the first three rows of each panel, each cell reports the point estimate of the $LQTE(\tau)$. Hypotheses tests are only carried out for those τ 's for which the estimate of $LQTE(\tau) \neq 0$.

all semesters. Secondly, we find an increase of approximately 20 days of employment for the median of the third semester of our sample. Beyond the point estimate, in the supplementary document, we show that the confidence intervals for all parameters of all quantiles include the zero. Thereby, we cannot reject that this effect on the median is simply due to chance. Still, we can reject the hypothesis of no effect at any quartile in the case of this third semester⁴. The point estimate decreases in the next semester and we cannot reject the treatment non-significance hypothesis at this period. In conclusion, all this evidence makes us argue that there was not a persistent effect of D_1 on the distribution of days worked.

Concerning the second treatment (D_2), we find increases of approximately two months of employment at the median of the distribution for the second semester. This point estimate is of remarkable size considering that the range of the outcome is $[0, 184]$ days. Unfortunately, the estimation is not very precise and, as shown in the supplementary document, its confidence interval includes the zero. Therefore, we cannot reject that the observed difference of quantiles is due to the sampling. The median effects contrast with point estimates around zero both for the first and the third quartiles of the distribution in almost every semester. Considering the statistical significance

⁴This positive effect at the median contrasts with a smaller and not statistically significant effect at the mean of the same quarter. The point estimates for the first and third quartile derived from a bias-corrected estimator are slightly negative, but not statistically significant. The same occurs for more extreme quantiles. Unfortunately, this less-biased estimation involves more imprecision. We argue that the negative impacts we found would explain the median-mean difference. These additional robustness checks overall support the main results presented in this article and are available upon request.

of our results, we cannot reject the hypothesis of no effect in any quartile for the number of days worked if we establish the conventional significance level (at $\alpha = 0.05$). Thereby, in our sample, the most relevant distributional impact of treatment D_2 is a 58 days effect at the median during the second semester. Nevertheless, this heterogeneous effect fades out along time.

Since assigning more money to participants does not overall generate a positive *LATE* on the employment duration, maybe it does on the quality of the employments. Unfortunately, we can only estimate the impact on the whole interval of two years after treatment started. Figure 4 offers visual evidence on this issue regarding two types of employment contracts as possible outcomes of the treatments: an open-ended contract (OEC) and a fixed-term contract longer than 12 months (FTCm12). Plots for contracts of shorter duration are available in the supplementary document. The main message from the plots is that there is no clear jump at the cutoff, so there seems to be no effect on employment quality. We can appreciate small discontinuities for the OEC outcome for both treatments, but this conclusion would require a more fine-grained analysis with numeric estimates.

Figure 4: RD plots with local proportions for employment quality variables.

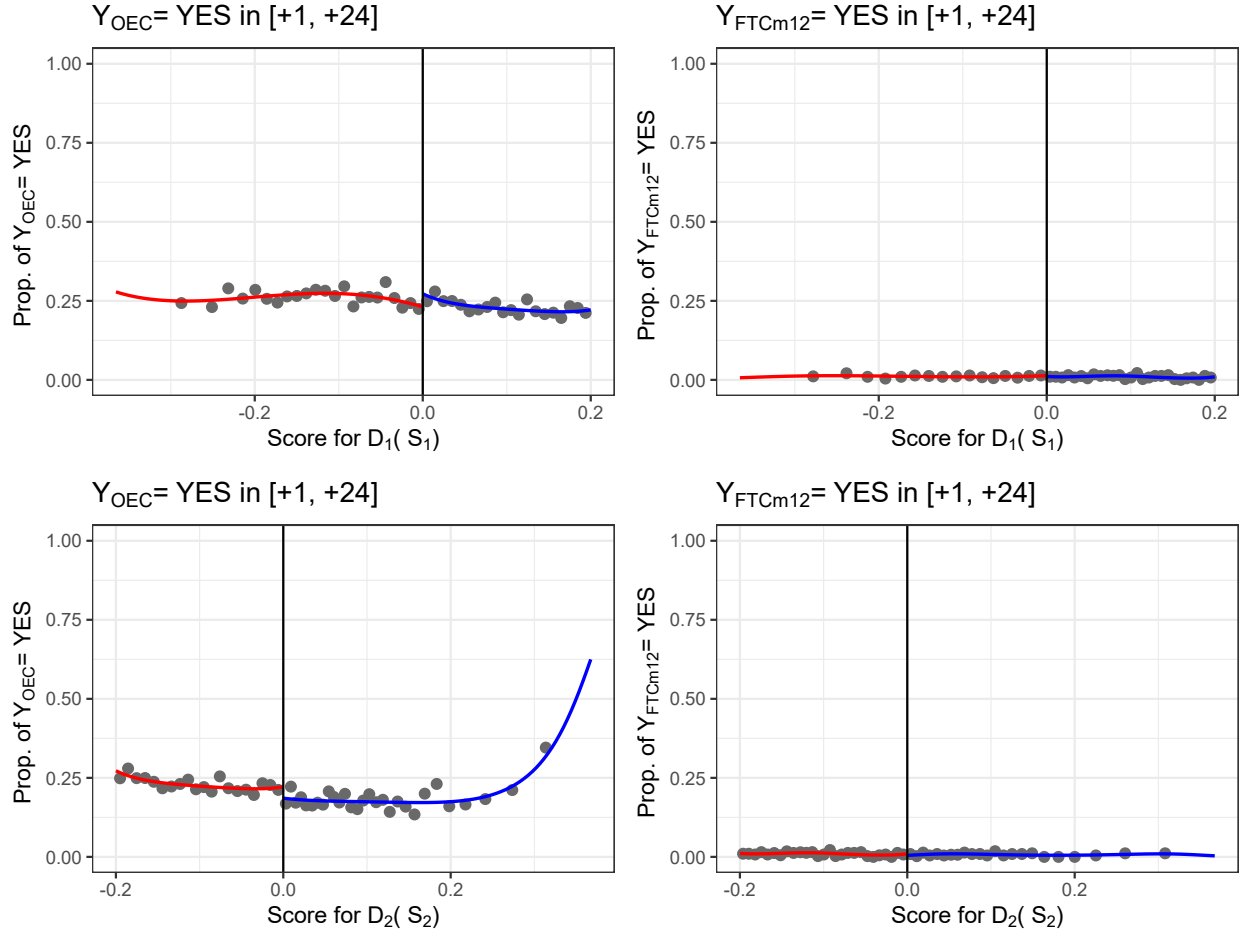


Table 5 presents the formal estimation of the impact of AXL on the probability of signing each

Table 5: Estimates of employment quality effects for each treatment and contract of the longest employment spell.

	Y_{OEC}	Y_{FTCm12}	Y_{FTC612}	Y_{FTC05}
D_1	0.019* (-0.001, 0.061)	0 (-0.01, 0.006)	0.018 (-0.009, 0.046)	-0.02 (-0.062, 0.01)
D_2	-0.031** (-0.062, -0.004)	0.002 (-0.004, 0.008)	-0.006 (-0.038, 0.015)	0.045*** (0.018, 0.092)

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each cell shows the point estimate of the *LATE* and 95 % confidence interval in brackets.

type of contract as the longest employment contract signed in two years. The columns collect both the contracts allowed (open-ended or OEC, long temporary or FTCm12 and intermediate temporary or FTC612) and not allowed (short temporary or FTC05) to receive the payment. The point estimates are overall not statistically significant at the conventional level ($\alpha = 0.05$) in both treatment comparisons. In the comparison of the high vs. intermediate endowments (D_2), we do find an effect of 4.5 p.p. on the probability of having a short temporary contract during the longest employment. At the same time, there is a decrease in the probability of having an open-ended contract of 3.1 p.p. in this treatment contrast. This might explain the increases at the median of the distribution that appeared for D_2 , because they were of low magnitude and only at the second semester. Such increase in short temporary contracts is interesting, because this is not a type of contract that guarantees the performance-based pay. One possible interpretation is that caseworkers thought that this short contract could facilitate getting a longer one that would get them access to the placement prize.

To examine the process of job search intermediation, we may compare the results about the effects of the participation in each group on two slightly different outcomes: the longest contract (Table 5) and the awarded contract (tbl-emplawards). While the former variable captures the most stable contract signed within two years from the start of the treatment, the latter variable refers to the contract linked to a placement prize (if any).

First, being in the intermediate-dose group (D_1) increases the probability of getting any of the awarded contracts. It seems that, in this case, a larger incentive for the provider triggers a larger effort to support the job search. Thereby, the treatment appears to have accelerated access to an employment to a (small) fraction of the participants. Interestingly, the point estimates among the three types of contracts are close and do not follow a linear function. If we remind the effects on the longest employment contract in two years presented in Table 5, we see that the active treatment has null effects on the probability of each type of contract. Thus, along time, either (a) the control group closes the gap with the active treatment group within two years or (b) the active treatment group is unable to retain the employment. In the latter case, policymakers could incorporate a job maintenance intervention involving meetings during the period of employment.

Second, we see that being in the high-dose group (D_2) has no effect on the awarded contract,

Table 6: Estimates of employment quality effects for each treatment and contract of the awarded employment spell.

	Y_{pOEC}	$Y_{pFTCm12}$	$Y_{pFTC612}$
D_1	0.022*** (0.008, 0.05)	0.01*** (0.005, 0.021)	0.048*** (0.019, 0.073)
D_2	0.008 (-0.019, 0.024)	0.001 (-0.011, 0.009)	0.019 (-0.007, 0.055)

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each cell shows the point estimate of the *LATE* and 95 % confidence interval in brackets.

but it does on the probability of having a short-term temporary job as the longest employment contract. On one side, we could interpret that people in this group required a longer timeline to adapt themselves to the labour market because they enter better occupations after treatment. This idea does not find support in our data, because the probability of getting a job with the occupation o during the longest employment contract barely changes among groups⁵. On the other side, we could interpret that caseworkers push treated jobseekers to accept less stable contracts with the hope of getting access to more stable contracts in the future. This would explain the null effects shown in Table 6 on the probability of signing any of the awarded contracts. However, as we can see in Table 5, participants in the high-dose group do not achieve a greater probability of signing a longer temporary contract and decrease their probability of getting a permanent contract.

4.3 Cream-skimming

One of the common rationales to establish different performance pays for each profile of workers is to discourage cream skimming of users, i.e. the selection of easy-to-place workers (Carter and Whitworth 2015). Policymakers anticipate that possibility and may include this design feature to prevent it, but there is no guarantee that the incentives are high enough to avoid this behaviour. In this section, we evaluate whether such discrimination exists under the current programme design. Considering that each provider might have different internal policies about the prioritization of users and that users self-select in a specific provider, we stratify the sample by provider for the subsequent analyses.

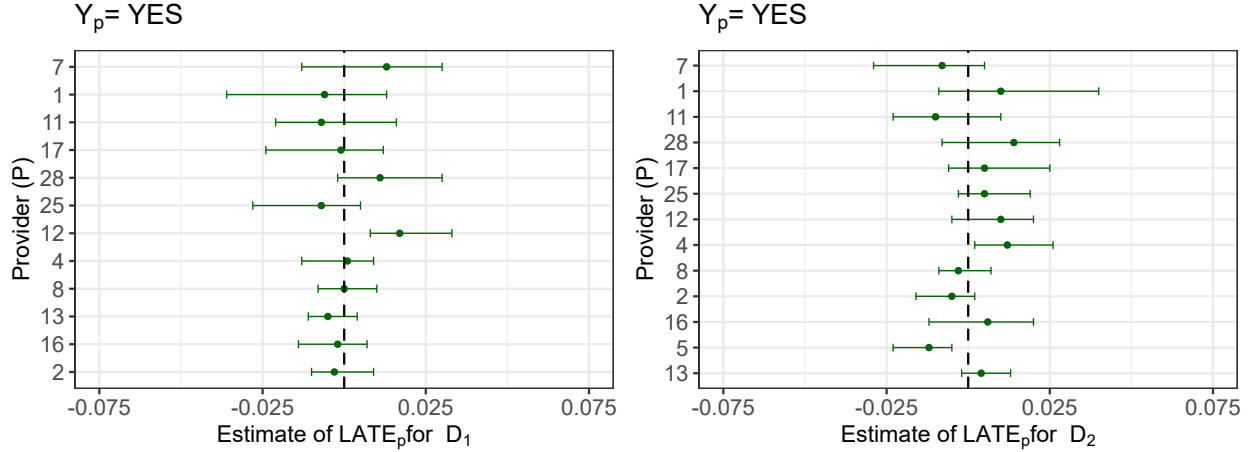
Figure 5 shows that there is almost no evidence of cream skimming, since we cannot reject that being in a less-employable treatment group changes the participation probability at the cutoff⁶. Departing from a theoretical model in which the untreated and the treated potential outcomes are monotonically related (Heckman et al. 2002), providers could interpret S (or D) as a proxy of the untreated potential outcome and select individuals based on that score. If cream skimming occurred, we would appreciate a jump in the regression discontinuity plots or, equivalently, a

⁵Results are available upon request

⁶To avoid small sample sizes, we only considered the set of providers that accounted for 75% of the sample (having previously sorted the providers from more to less users of AXL).

non-null $LATE$ on the participation probability when the endowment (the profile assigned with the score) is higher.

Figure 5: Estimates of the $LATE_j$ on the probability of participating in each provider. Each dot represents the point estimate and the bars show the 95 % confidence interval.



For both treatments, in general, we do not detect a statistically significant effect of being assigned to a group with larger endowment on the probability of participating with each provider. Interestingly, two of the three coefficients that are statistically significant are positive, which indicates that a greater endowment fostered the participation in certain providers. This indicates an “inverse cream skimming” in which providers select hard-to-place workers, something that could interest some governments in terms of fairness. The unique exception is provider 5 in D_2 , for which we estimate that a larger treatment endowment decreases the probability of participating in that provider at the threshold. In any case, this evidence of cream skimming appears incidental rather than systematic, because that provider only represents around 1 % of the sample. Therefore, on a positive note, the programme appears to have effectively deterred discrimination in the access to job search services.

4.4 Plausibility and robustness checks

We have performed two plausibility checks to analyse placebo outcomes and covariate balance. Then, we have carried out some robustness checks by varying the bandwidth, the kernel, or the polynomial degree. Here we summarize the main results of both exercises, while in the supplementary document the reader can find all the estimates.

The first plausibility check employs placebo outcomes to search evidence about the presence of an alternative programme using the same scoring variable to assign treatments. Leveraging the staggered implementation of the AXL treatments, we can estimate pseudo-treatment effects at the moment the programme was already implemented but certain individuals had not yet participated. Differences at these pretreatment periods (for individuals treated at semester $t \geq 2$) would inform on possible violations of $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp (\tilde{S}_i)$ for individuals treated at semester

$t = 1$. If we focus on the effects at the average, the point estimates are small and not statistically significant at the usual level. This would support the absence of an alternative treatment that might bias the estimation of the AXL effect due to a discontinuity in the untreated potential outcome functions. Additionally, the documents on the programme design inform that the score function was published in November 2017 and the mandate to create such function was passed in July 2017 ([Government of Veneto 2017a](#)). This finding goes in line with the idea of no alternative treatment at $\tilde{S}_i = c$ before AXL implementation, since other agents did not know –at least through open channels– the score function.

The second plausibility check, the covariate balance analyses, inform us about the comparability of active and control groups to bring evidence on the unconfoundedness condition. Attending the magnitude of the differences, only 1 out of the 52 variables/categories analysed (26 for each treatment comparison) presents an imbalance higher than the 0.1 recommended threshold derived from Stuart et al. (2013) and Austin (2009). Following Rosenbaum (2010), we also looked at the unstandardized mean difference to interpret the size of the problem. There is an average difference of one year of age between groups for D_1 , which corresponds to 0.13 standard deviations. Considering the low standardized difference and the practical significance of one year for the employment probability during most of the working life, we argue that the confounding effect of the tiny difference of age is negligible for our conclusions⁷.

The robustness checks deal with certain changes in the modelling decisions and their effects on the conclusions. The first set of analyses preserve the same population of interest. Regarding the *LATE* results, we have varied two modelling options: bandwidth selector (MSE-optimal but allowing different bandwidths at each side of the cutoff and a slightly lower common MSE-optimal bandwidth) and polynomial degree (1 or 2). Then, we have compared the estimates presented in Section 4.2 with the alternative results. Our results on the *LATE* are quite robust to the different specifications both in magnitude and direction. With the alternative bandwidths, the results are almost identical up to the differences in statistical significance that might be explained by the distinct power of tests with different sample sizes. Concerning the $LQTE(\tau)$ and $LATE_j$ results, the robustness analyses are less established because the estimators have been recently proposed. For the $LQTE(\tau)$, following Qu and Yoon (2019), we have re-estimated the models with a new bandwidth selector by choosing the second most conservative bandwidth, rather than the most conservative (i.e., the smallest). For both comparisons, the results for the median are qualitatively equal with only small differences in quantitative terms. In all cases the conclusions are preserved. For the $LATE_j$, we have applied a different bandwidth for each category of the outcome by transforming the multinomial model to a set of binomial models. The qualitative results are preserved and the quantitative estimates are almost equal, so the conclusions are robust.

⁷As an additional robustness check, we rerun the *LATE* analyses controlling for the residual imbalance in those covariates with differences of at least 5 p.p. (or 0.05 standard deviations) and conclusions do not change. Results are available upon request.

5 Conclusion

In this article, we have studied how the intensity of job search services influences its employment effects. To do so, we have leveraged the discontinuity design of an innovative policy that combines two design features for the private provision of public services: vouchers (for job search assistance) and performance-based payments (for job search intermediation). In this policy, three treatment groups shared the access to adult training, but had different available doses of other components: group *A* (with 7 hours of assistance and 1,500 € for intermediation), group *B* (with 13 hours of assistance and 2,000 € for intermediation), and group *C* (with 27 hours of assistance and 3,000 € for intermediation).

The results indicate that the local effect on the average of days worked was null for all semesters in the first comparison (*B* vs. *A*) and for three out of four semesters in the second comparison (*C* vs. *B*). In this last case, we did detect a small but positive effect during the second semester that later vanishes. We did not find effects at the first and third quartile of the distribution of days worked, but some changes appeared at the median. For the intermediate-low comparison, we detected an approximate increase of around two months at the median of the third semester. The high-intermediate comparison presented a rise of twenty days at the second semester after treatment started. Nonetheless, both effects disappear at the following semesters. Lastly, it seems that the different treatment intensities generated certain impacts on the employment quality, although these contracts did not trigger longer employment episodes. We found a small positive effect on the probability of having short temporary contracts as the longest employment for the high-intermediate contrast, whereas the intermediate-low contrast presented an increase of permanent contracts during programme implementation. We also found two important results regarding implementation. First, the mechanism that might explain these minor employment impacts did not seem to be job search assistance due to voucher non-redemption, but job search intermediation. Second, we did not detect evidence of cream-skimming or discrimination of hard-to-place jobseekers in the access to job search services under the different performant payments, which suggests that it is a promising design feature.

Our findings have several policy implications. Regarding job search assistance, if policymakers wish to ensure that less-employable individuals receive more treatment, we have seen that these vouchers do not ensure it. In following versions of AXL, vouchers might be designed as those usually available in education policies. In these cases, the individual may choose the provider, but not the number of hours of education. Regarding job search intermediation, the payment differentials studied here do not appear sufficient to generate sustained employment gains. Nonetheless, it is possible that the differences between the payments associated with each group were not high or directed enough to cause the desired increase in caseworkers' effort.

The limitations of our research are mainly two. Firstly, regression discontinuity only identifies the effects for individuals just at the threshold of the score. If the impact of the programme was heterogeneous as a function of the running variable, our results would not be generalizable to individuals far from the cutoffs. Secondly, the flexible expenditure of the voucher allowed that

different endowments did not materialize in remarkably different amounts of assistance received. On one side, this might induce us to think the results exclusively as the effects of different payment schemes for intermediation. On the other side, we could have a more complete view and interpret the findings as intention-to-treat effects. This last approach understands the lack of voucher redemption as a rejection of the assumptions behind the programme design formulated by the policymakers.

Future research might reinforce either the impact or the implementation analyses of job search services. On the implementation, it would be necessary to investigate in greater depth the specific actions carried out as job search assistance. As remarked by Hernán and Robins (2023), it is required to define in a precise way the interventions we want to analyze. Nowadays, the public employment service lacks detailed information on the actions carried out by each specific provider. On the impact, it would be interesting to consider control groups made up by individuals that received no treatment. Although identification would require harder assumptions, the results would complement the view on the worthiness of the programme. Lastly, aggregate effects seem quite small, so cost-benefit analyses might help to decide the pertinence of this programme.

In sum, our work does not find support to reserve more money to treatments of different intensity since it does not seem to generate lasting effects on the treated people. Future policy designs might consider devoting such money to treating more jobseekers instead of increasing the treatment endowment of each participant.

Statements and declarations

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Competing interests. The author has no competing interests to declare that are relevant to the content of this article.

Data and computer code availability. The data that support the findings of this study are available from Veneto Lavoro, but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Please, send an e-mail to Veneto Lavoro (osservatorio.mdl@venetolavoro.it) for further details about how other researchers can obtain the data and the procedure to request the data. All the information needed to proceed from the raw data to the results of the paper (including code) is available in a [GitHub repository](#).

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